DEVISHI KESAR

Email: dkesar20@gmail.com • LinkedIn • Github Profile • Website

EDUCATION

German Cancer Research Center(DKFZ) — ***Heidelberg, Germany***
Doctor of Natural Science — *University of Heidelberg*
May 2027

Georgia Institute of Technology — ***Atlanta, GA, USA***
Master in Bioinformatics
December 2020

Indraprastha Institute of Technology Delhi — ***Delhi, India***
Bachelor of Technology (Honors) in Computer Science and Engineering
Minor in Computational Biology
August 2019

PUBLICATIONS

• Jen, J., He, Y., Frederick, D. T., Marzoratti, A. M., Thakar, T., Chen, H., Zhang, A., Acosta, A., Li, G., Feng, B., Chokshi, M., Jung, N. Y., Bayyat, Z., Yang, Z., Oliver, S., Balasooriya, E. R., Wienand, K., Stokes, S., Inam, H., Pezzulo, D., Rodriguez, D. J., Young, M. J., Guo, A., **Kesar, D.**, Li, R., Ito, T., ... Sellers, W. R. PARADIGM: A PARAlog DIGenic Matrix for Mapping Synthetic Lethality and Therapeutic Vulnerabilities in Cancer. *under review*

• van Tienen, L. M., Bayoumi, S., Muneeruddin, K., Leymarie, N., Popa, A., Shekhar, M., Mueller, M., Li, R., Zak, K. M., Chilukuri, H., Kornfilt, D. J. P., Atack, T. C., **Kesar, D.**, Bian, Y., Shaw, K. L., Jandova, Z., Trollmann, P., Geist, L., Stolt-Bergner, P., ... Sellers, W. R. Systematic cysteine scanning identifies a druggable pocket in oncogenic KRAS (2026). *Cell Chemical Biology*

• Knoll, N., Masser, S., Bordas, B., Ebright, R. Y., Li, G., **Kesar, D.**, Destefanis, E., Kania, N., Rodriguez, D. J., Jen, J., Zagar, S. E., Mensah, C., Chen, Z., Moffitt, S. J., Enakireru, E. M., He, Y., Feng, B., Chokshi, M. K., Jin, C. Y., Raghavan, S., Sellers, W.R., Mulvaney, K. M., CRISPR-drug combinatorial screening identifies effective combination treatments for MTAP-deleted cancer (2025). *Cancer Research*

• Shirole, N., **Kesar, D.**, Lee, Y., Goodale, A., Syamala, S., Kukreja, S., Li, R., Qiu, X., Yu, W., Goldman, S., Cejas, P., Long, H., Adelman, K., Doench, J., Sellers, W.R. and Kaelin, W., Sustaining Cyclin D1 Expression Renders Kidney Cancer Cells HIF2-Independent (2025). *Cancer Discovery*

• Chaturantabut, S., Oliver, S., Frederick, D.T., Kim, J., Robinson, F.P., Sinopoli, A., Song, T.Y., He, Y., Chang, Y.C., Rodriguez, D.J., Chang, L., **Kesar, D.**, Ching, M., Dzvurumi, R., Atari, A., Tseng, Y.Y., Bardeesy, N., Sellers, W.R., Identification of potent biparatopic antibodies targeting FGFR2 fusion driven cholangiocarcinoma (2025). *The Journal of Clinical Investigation*

• Chang, L., Jung, N.Y., Atari, A., Rodriguez, D.J., **Kesar, D.**, Song, T.Y., Rees, M.G., Ronan, M., Li, R., Ruiz, P., Chaturantabut, S., Ito, T., Tienen, L.M., Tseng, Y.Y., Roth, J., & Sellers, W.R., Systematic profiling of conditional pathway activation identifies context-dependent synthetic lethalities (2023). *Nature Genetics*

• Li, R., Klingbeil, O., Monducci, D., Young, M. J., Rodriguez, D. J., Bayyat, Z., Dempster, J. M., **Kesar, D.**, Yang, X.,Zamanighomi, M., Vakoc, C. R., Ito, T., & Sellers, W. R., Comparative optimization of combinatorial CRISPR screens (2022). *Nature Communications*

• Mulvaney, K. M., Blomquist, C., Acharya, N., Li, R., Ranaghan, M. J., O'Keefe, M., Rodriguez, D. J., Young, M. J., **Kesar, D.**, Pal, D., Stokes, M., Nelson, A. J., Jain, S. S., Yang, A., Mullin-Bernstein, Z., Columbus, J., Bozal, F. K., Skepner, A., & Sellers, W. R., Molecular basis for substrate recruitment to the PRMT5 methylosome (2021). *Molecular Cell.*

• Mason, M.J., Schinke, C., Eng, C.L.P. et al. Multiple Myeloma DREAM Challenge reveals epigenetic regulator PHF19 as marker of aggressive disease (2020). *Leukemia* - **member of DREAM consortium**

• Yang, et al. Assessment of the Limits of Predictability of Protein and Phosphorylation Levels in Cancer (2020). *Cell Systems* - **member of NCI-CPTAC-DREAM consortium**

RESEARCH EXPERIENCE

Pediatric glioma research lab, DKFZ
Heidelberg, Germany
Graduate student *October 2023 – Present*

- Processed ChIP-seq to characterize TF binding and histone-mark signatures in pediatric gliomas
- Conducted TF motif analysis to prioritize regulatory elements driving subtype programs
- Constructed enhancer–TF–gene networks to uncover candidate biomarkers across glioma subtypes
- Evaluated multi-omic integration approaches to link chromatin features to gene expression patterns
- Performed single-cell and Xenium spatial analysis to map biomarker expression and infer developmental bottlenecks

Sellers Lab, Broad Institute of MIT and Harvard
Cambridge, MA, USA
Associate Computational Biologist *April 2021 – September 2023*

- Processed and analyzed data for CRISPR paralog redundancy screens to find actionable biomarkers
- Executed antibody docking and structure prediction for antibodies using modeling softwares
- Performed ChIPseq data analysis to look at TF motif enrichment and single nucleotide resolution cut profile
- Detected alternate splicing events helping suggest PBM interaction requirement for PRMT5-dependent splicing events
- Analyzed data from mass spectrometry experiments conducted to identify novel E3 substrates

Georgia Institute of Technology Atlanta, GA, USA *Graduate Research Assistant* *August 2019- December 2020*

- Benchmarked variant calling pipelines by comparison of performance of mappers and variant callers
- Cataloged genomic variants from Colombian populations and characterized their predicted functional impact
- Identified single variant disease associations using best practices from the American College of Medical Genetics
- Evaluated distribution of health-related genetic variants among populations from Colombia in light of their ancestry

EMD Serono Billerica, MA, USA *Bioinformatics Summer Intern* *May 2020 – July 2020*

- Developed Rshiny app to share isolated immune cell RNAseq from bulk breast cancer tumors with biological experts
- Discovered novel myeloid drug targets by leveraging differential expression and gene signature analysis
- Drove bioinformatics discovery by frequently presenting results to key stakeholders across multiple departments
- Identified actionable biomarkers supporting therapeutic remediation of immuno-oncology and autoimmune disease

INVITED TALKS

- *Deciphering the cell of origin in pediatric gliomas* 10x Genomics World Tour, Heidelberg, Germany. April 2026.
- *Pediatric Tumors and the Search for Neoantigens*, FBRI DC Computational day, Washington D.C., USA. August 2025.

POSTER PRESENTATIONS

- **Kesar, D.**, Keck, M.K., Joshi, P., Okonechnikov, K., Autry, R.J., T.W. Jones, D.T.W., Deciphering pediatric glioma subtypes—Super-enhancer dynamics and (epi)genomic insights into cell of origin, Systems Biology, CSHL 2026
- **Kesar, D.**, Keck, M.K., Joshi, P., Okonechnikov, K., Autry, R.J., T.W. Jones, D.T.W., Integrated (epi)genomic atlas for identification of biomarkers in pediatric gliomas, AACR 2025
- **Kesar, D.**, The discovery of context-specific paralog dependencies in cancer, Broad Retreat 2022
- **Kesar, D.**, Colombian Genome Diversity Project, GT@SC20 HPC Poster Show Proceedings

TECHNICAL SKILLS

Programming and Computational Skills: Python, Bash, R, Java, Scala, Regex, SQL, MATLAB
Tools and Platforms: Git, Hive, Cytoscape, RStudio, PyMol, Databricks, AWS, Spark, Docker, Google Cloud Services

AWARDS

- Computational Biology Research Award – Spring 2020 and Fall 2020 – Georgia Tech
- Dean Academic Excellence Award – IIIT Delhi 2018
- RECOMB/ISCB Conference Travel Grant 2017

LEADERSHIP AND VOLUNTEER EXPERIENCE

- Member, Social events team, DKFZ, Germany, 2023-25
- Organizing team, Broad Retreat 2022, Broad Institute, USA, 2021-22
- Volunteer, Charles river cleanup, The Charles River Conservancy, MA, USA, 2021-22
- Volunteer, GP Covid testing lab, Broad Institute, MA, USA, 2022